

Analysis of Noise removal in ECG signal using Symlet wavelet

¹ B.Vijayakumari, ²J.Ganga Devi, ³M.Indhu Mathi.

¹ Associate Professor, Kamaraj College of Engineering and Technology, Virudhunagar, India.

² Assistant Professor, Kamaraj College of Engineering and Technology, Virudhunagar, India .

³ PG Student, M.E (CNW), Kamaraj College of Engineering and Technology, Virudhunagar, India.

indhuece93@gmail.com

gangamahitha@gmail.com

Abstract

Nowadays Cardio vascular diseases becomes big threat to the life span of human beings. ECG is one of the best method to give clear information about cardiac arrhythmia. ECG signal is contaminated by various noises of various stages. To remove noise from the ECG signal, dwt based symlet wavelet function is used. In this work, the ECG signals are taken from the MIT-BIH arrhythmia database. The major sources of noise like color noise, real muscle artifact, electrode motion and baseline wander noises are taken from MIT-BIH noise stress test database. This methodology involves four steps. They are decomposition, thresholding, denoising and calculate SNR values for ECG signals. DWT based Symlet wavelet function is used to decompose and denoising the ECG signal. After decomposing, thresholding technique is applied to detailed coefficient, to obtain threshold signal. Denoised signal is obtained from threshold signal. Finally the SNR value is calculated for denoised signal. The simulation results show that the proposed work is able to reduce noise from the noisy ECG signals and it also reliable even the signal condition is poor.

such as power line interference, baseline wander, electrode motion, muscle artifact noise. Color noise is occur in telemedical application when the channel condition is poor.



Fig.1.ECG signal

Keywords: Symlet wavelet transform, thresholding , signal denoising

I.Introduction

Nowadays, cardiovascular diseases (CVDs) are the leading causes of death and disability in the world. Heart failure occurs when heart's pumping power is weaker than normal and hence movement of blood becomes slow and increase the pressure. Among several measuring devices, ECG is considered as best measuring tool for measuring arrhythmia by placing electrodes in patient's body. ECG is the process of recording the electrical activity of the heart. Figure1 shows that representation of ECG signal. ECG waves are labeled using the letters P, QRS, T and U. ECG signal gives useful information about patient's heart. Some noises may present in that signal during measurement and transmission. For clear prediction of arrhythmias, those noises must be reduced. The noises

II Materials

MIT-BIH arrhythmia database

MIT-BIH arrhythmia database was also used to obtain real ECG signal. It consists of 48 records, obtained from the Beth Israel Hospital in Boston. Each record having sampling frequency of 360 Hz. 'Header' files gives the information about the patient's age, sex, and medications.

MIT- BIH noise stress test database

This database includes the different types of noisy signal which are corrupted the ECG signal during measurement. These databases includes different types of noise such as baseline wander noise, electrode motion noise, muscle artifact noise etc.

Color noise

Color noise is low frequency noise which is occurred by the poor channel condition in telemedical applications [1].

III Methods of this work

1.Symlet wavelet transform

The modified version of Daubechies wavelet is called Symlet wavelet. The scaling function and wavelet function of symlet wavelet looks like the shape of ECG signal and it also provide better denoising of noisy ECG signal. Here the wavelet order 8 and decomposition level 5 is choosen [2]. Figure.2 shows the flow chart of this work .

2.Thresholding techniques

The signals carry a large amount of useful information which is difficult to find, the smaller amount of wavelet coefficients reveals the information of signal. It is found by using thresholding methods. Here three types of thresholding method is used [3].

A .Universal thresholding

This is the basic and general thresholding method. It easily find the accurate information of the signal. The threshold value is given in equation as

$$\text{universal} = \sigma_i \sqrt{2 \log(I)} \text{ -----(1)}$$

where I is the number of data samples [4] [5].

B. Minimax Thresholding

Minimax is another global thresholding method.This method finds threshold using Minimax principle [6] [7] . The threshold is given by:

$$\text{Minimax} = \sigma_i (0.3936 + 0.1829 \log_2(I)) \text{ -----(2)}$$

σ_i is the standard deviation of noise and I is the number of samples. The minimax estimator which can provide the minimum value from set of wavelet coefficients and give low frequency signal. The smaller wavelet coefficients of the lower frequency signal give information about noise in a noisy signal. Minimax uses a fixed threshold to estimate an accurate information about patient's heart beat.

C.Heursure Thresholding

Threshold is selected using a combination of Sqrtwolog and Rigrsure methods. If the signal condition is poor the combination of both thresholdings is used and gives the better result [8].

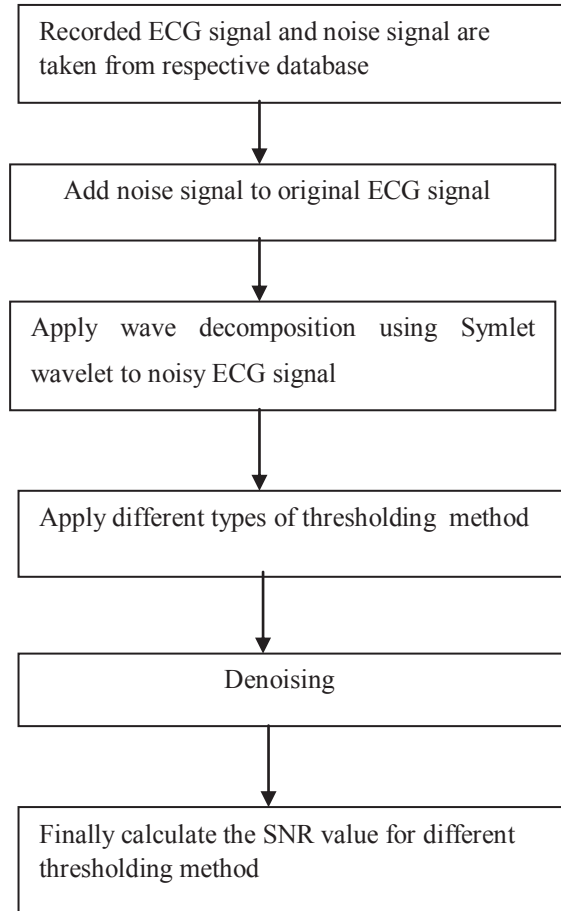


Fig.2. Flow Chart

3.Denoising steps

The signal denoising involves following steps [11] [12],

- Decomposing the ECG signal.
- Apply different types of wavelet methods
- Thresholding is applied to detail coefficients from decomposing operation.
- Either soft or hard thresholding is applied to detail coefficients .
- Denoising the ECG signal.
- Estimation of SNR value.

4.SNR Calculation

Signal to noise ratio which gives the information about the quality of the signal. SNR formulas given as follows,

$$SNR = 10 \log \left[\frac{\sum d^2(n)}{\sum (d(n) - k(n))^2} \right] \text{---(3)}$$

Where $k(n)$ is the input signal and $d(n)$ is the denoised signal [13].

IV Results and Discussion

This chapter focus on the outputs obtained at each step as mentioned in the previous chapters. The proposed algorithm is evaluated with the database consist of ECG signals. By referring the given flow chart in Fig.2 the first step is to taken the ECG signals and noise ECG signal from MIT-BIH Arrhythmia Database , MIT-BIH noise stress test Database as shown in following figures 1, 2, 3, 4, 5.

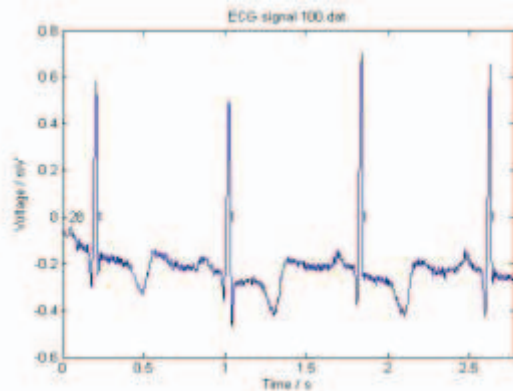


Fig.3 Input signal

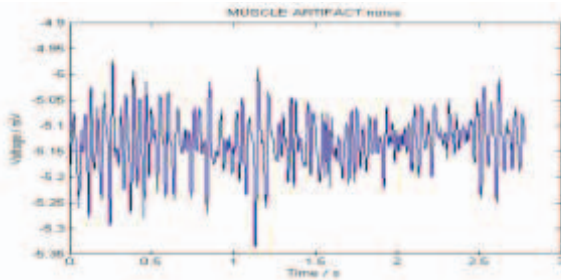


Fig.4 Muscle artifact noise

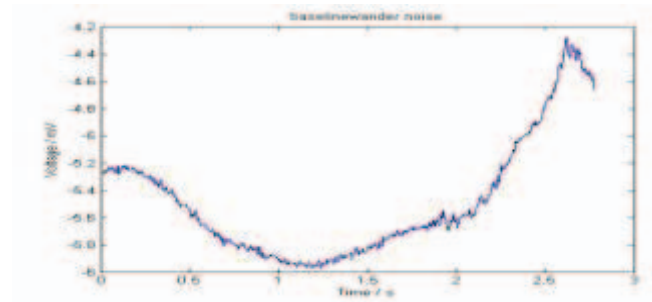


Fig.5 Baseline wander noise

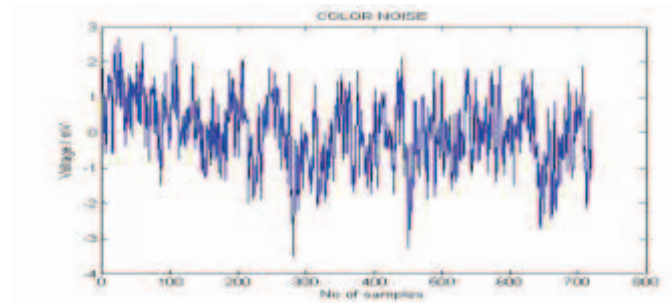


Fig.6 Color noise

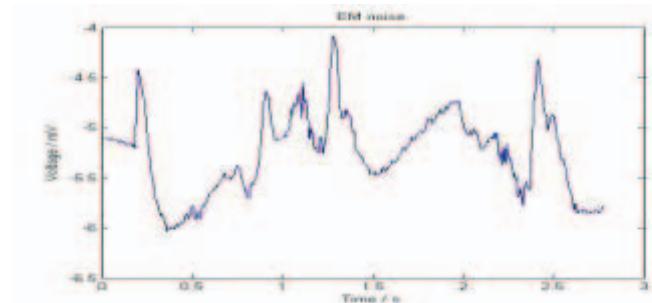


Fig.7 Electrode movement noise

The input signal is taken from the MIT-BIH Arrhythmia database.

1. Results obtained for noisy signals

In step 2, the above mention noise signals are added to the input ECG signal to get noisy ECG signal and the following figures are as follows,



Fig.8 BW noise added to input ECG signal

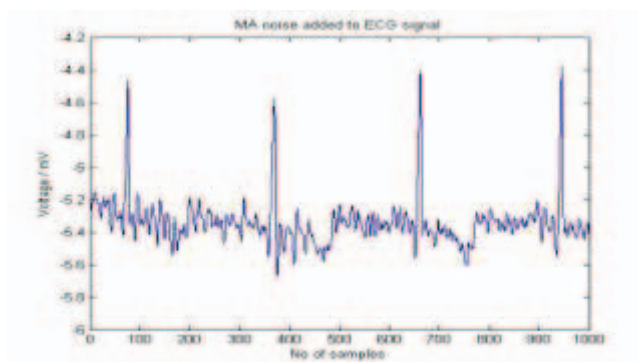


Fig.9 MA noise added to input ECG signal

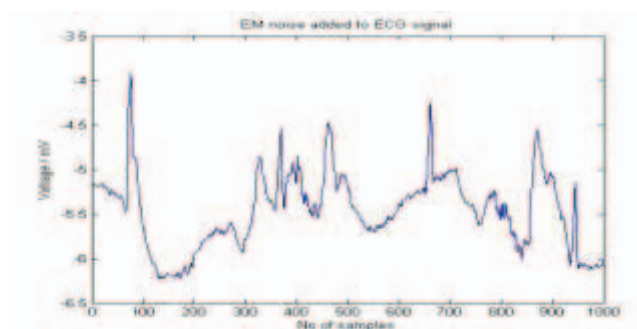


Fig.10 EM noise added to input ECG signal

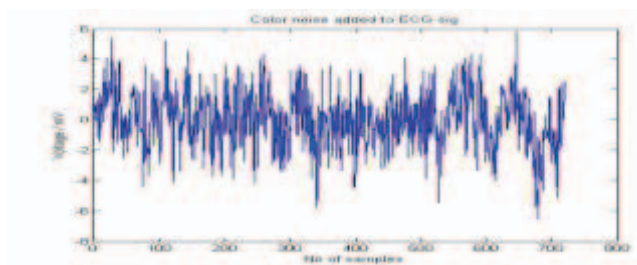


Fig.11 Color noise added to input ECG signal

2. Results obtained for denoise signal

The obtained noisy ECG signals are decomposed by symlet wavelet in step 3. For removing noise from the noisy signal different types of thresholding such as universal, minimax and heursure methods are applied in step 4 and the following figures shows the denoised signals output.

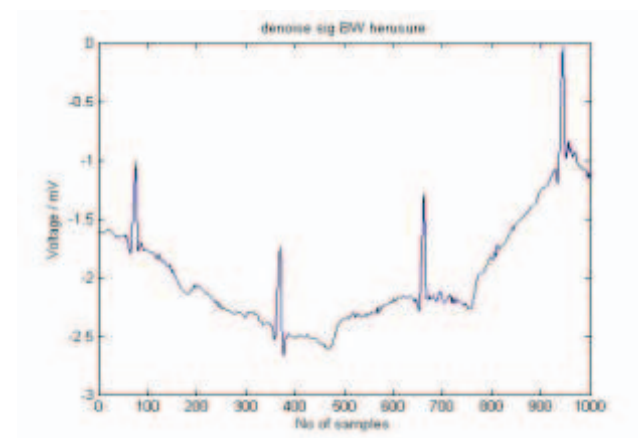


Fig.12 BW denoise signal

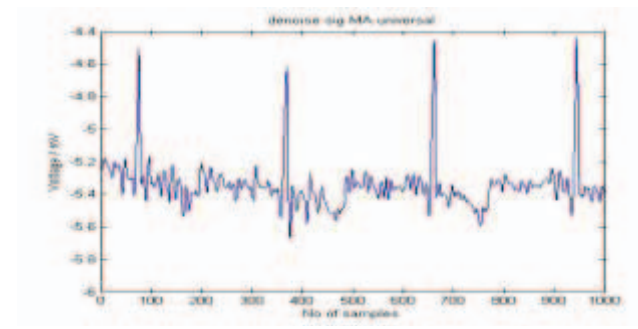


Fig.13 MA denoise signal

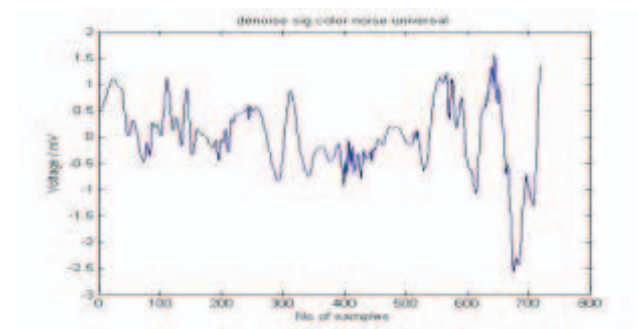


Fig.14 color noise denoised signal

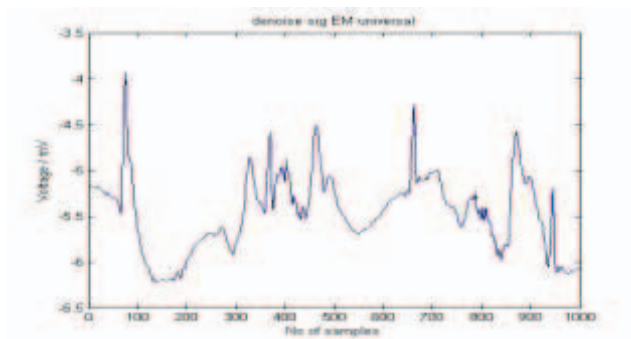


Fig.15 EM denoise signal

3. Results obtained for SNR improvement values

Finally the SNR values are calculated for different thresholding methods as shown in following tables 1,2,3,4,5,6 .

Thresholding Method	SNR improvement values			
	BW noise	MA noise	EM noise	color noise
Universal	0.0021	0.3109	0.3113	0.7891
minimax	0.0018	0.5470	0.5482	0.9245
heursure	0.0079	1.1234	0.0078	1.0678

Table 1 SNR values for Record No.100

Thresholding Method	SNR improvement values			
	BW noise	MA noise	EM noise	color noise
Universal	0.0043	0.5592	0.5584	0.0921
minimax	0.0096	0.9450	0.9733	0.7651
heursure	0.0139	1.9683	0.0154	1.6821

Table 2 SNR values for Record No.101

Thresholding Method	SNR improvement values			
	BW noise	MA noise	EM noise	color noise
Universal	0.0651	0.0196	0.0196	0.0560
minimax	0.0601	0.0346	0.0346	0.9217
heursure	0.0665	0.0722	0.0670	1.4735

Table 3 SNR values for Record No.102

Thresholding Method	SNR improvement values			
	BW noise	MA noise	EM noise	color noise
Universal	0.0013	0.1382	0.1380	0.0548
minimax	0.0021	0.2454	0.2450	0.9543
heursure	0.0034	0.5180	0.0034	1.7632

Table 4 SNR values for Record No.103

Thresholding Method	SNR improvement values			
	BW noise	MA noise	EM noise	color noise
Universal	0.0057	0.7114	0.7100	0.0560
minimax	0.0101	1.2411	1.2381	0.9214
heursure	0.0184	2.5002	0.0204	1.4735

Table 5 SNR values for Record No.201

There are six recorded ECG signals are taken from the MIT-BIH arrhythmia database and calculate the SNR improvement values and analysis which thresholding gives the high SNR values.

Thresholding Method	SNR improvement values			
	BW noise	MA noise	EM noise	color noise
Universal	0.0038	0.5177	0.5164	0.7891
minimax	0.0063	0.9273	0.9235	0.9245
heursure	0.0134	1.9961	0.0150	1.0678

Table 6 SNR values for Record No.202

Inferences from the above tables indicates that the heursure threshold method yields comparatively better results for SNR values. But the EM noise only having high SNR at minimax thresholding method as shown in Table 6.

VI Conculsion

This Project proposes a efficient method to remove noise from ECG Signal. First the noise signals are added to ECG signal, then the noisy ECG signal is decomposed using symlet wavelet. By decomposing, the detail and approximation coefficients are obtained. Then detailed coefficient is applied to different types of thresholding technique, to obtain threshold signal. Finally denoise the threshold signal and SNR values are calculated for both realistic noises and color noise. This process is taken for all above mentioned noises. Compare to other thresholding method the heursure method gives the better results. This work can be further expanded to various performance evaluation such as SNR improvement value, mean square error, root mean square error and percent root mean square difference.

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